

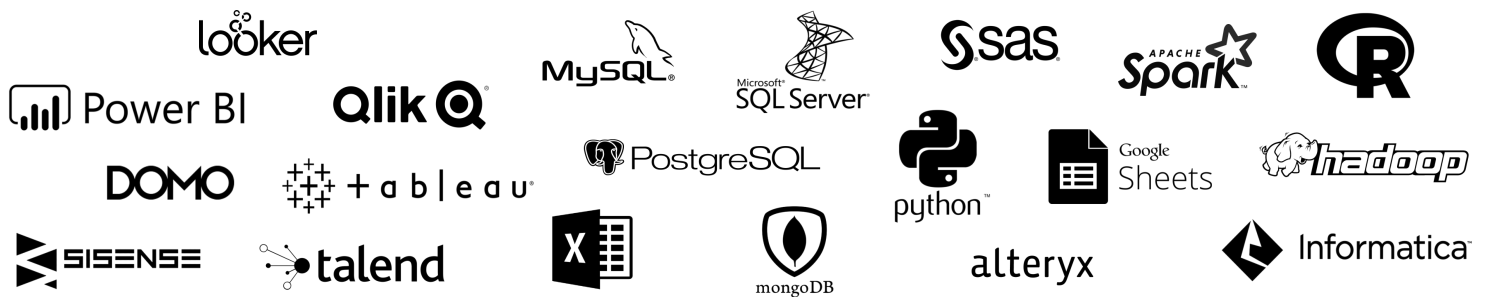
THE ANALYST TOOLKIT

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THE ANALYST TOOLKIT

TOOLS MATTER, BUT **SKILLS MATTER MORE**

When it comes to analytics tools, **don't be afraid to specialize**; focus on building deep skills and expertise in a few key areas, rather than trying to master it all



You have permission to **NOT** learn all of these tools

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THE ANALYST TOOLKIT

BI Platforms

Database Tools

Data Prep/ETL
Tools

Spreadsheet
Tools

Programming
Languages

BI PLATFORMS



Power BI



+ a b l e a u

Qlik

SISENSE

Self-service tools designed to support the entire business intelligence workflow, including data prep, modeling, analysis, visualization and administration



INDUSTRY
LEADERS*

- Power BI
- Tableau
- Qlik



TYPICAL
USER ROLES

- Business Intelligence Analyst
- Data Analyst
- Data Visualization Specialist



COMMON
USE CASES

- Performance dashboards
- ETL & data prep
- Ad hoc data visualization
- Report administration



PRO TIP: Rather than trying to do everything at once, focus on going deep with one platform first; the skills you build will be highly transferrable if you need to switch

* Other popular BI platforms include Sisense, Domo, Google (Looker), and MicroStrategy

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THE ANALYST TOOLKIT

BI Platforms

Database Tools

Data Prep/ETL
Tools

Spreadsheet
Tools

Programming
Languages

DATABASE TOOLS

MySQL

SQL Server

PostgreSQL

mongoDB

Used for creating, managing, and querying information stored in relational database management systems (RDMS)



INDUSTRY
LEADERS*

- MySQL
- Microsoft SQL Server
- PostgreSQL



TYPICAL
USER ROLES

- Database Administrator
- Business Intelligence Analyst
- Data Scientist



COMMON
USE CASES

- Storing large datasets
- Ad hoc database analysis
- Producing custom tables or views for analysis and visualization



PRO TIP: There are many “flavors” of SQL database tools, but they are all built on the same query language and universal standards (with slight variations in syntax)

* If you are looking for cloud database tools, consider tools like Amazon RDS/Redshift, Azure, MongoDB, etc.

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THE ANALYST TOOLKIT

BI Platforms

Database Tools

Data Prep/ETL Tools

Spreadsheet Tools

Programming Languages

DATA PREP/ETL TOOLS



Informatica

alteryx

Used for extracting, cleaning, transforming, and loading data from disparate sources into a centralized location for analysis



INDUSTRY LEADERS*

- Talend/Informatica (cloud)
- Alteryx
- Tableau Prep
- Power BI (Power Query)



TYPICAL USER ROLES

- Data Engineer
- Database Administrator
- Business Intelligence Analyst



COMMON USE CASES

- Building data pipelines for analysis
- Developing an automated ETL process for ongoing reporting
- Blending data from multiple sources into a central database or BI platform



PRO TIP: Tableau Prep and Power Query are both excellent ETL tools, and are directly integrated with tools you're likely already using (Tableau, Power BI & Excel)

* While these are popular options, you can also find specialized ETL tools designed specifically for real-time/streaming, unstructured data, etc.

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THE ANALYST TOOLKIT

BI Platforms

Database Tools

Data Prep/ETL Tools

Spreadsheet Tools

Programming Languages

SPREADSHEET TOOLS



Used for creating, managing, modeling and analyzing structured data stored in rows and columns



INDUSTRY LEADERS*

- Microsoft Excel
- Google Sheets



TYPICAL USER ROLES

- Data Analyst
- Business Intelligence Analyst
- Financial Analyst
- Anyone working with data



COMMON USE CASES

- Ad hoc analysis & visualization
- Financial modeling
- Forecasting & optimization
- Performance reporting



PRO TIP: Excel might just be the most versatile tool in the analytics stack, and can be a great solution for anything from ad hoc analysis to complex data modeling

* There are other spreadsheet tools on the market (LibreOffice/Calc, Zoho, etc.), but Excel and Google Sheets are in a league of their own

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THE ANALYST TOOLKIT

BI Platforms

Database Tools

Data Prep/ETL
Tools

Spreadsheet
Tools

Programming
Languages

PROGRAMMING LANGUAGES



Coding languages and packages commonly used for statistical analysis, machine learning and data science



INDUSTRY LEADERS*

- Python
- R
- SAS
- SQL



TYPICAL USER ROLES

- Data Scientist
- Machine Learning Engineer
- Data Engineer



COMMON USE CASES

- Training machine learning models
- Analyzing massive datasets
- Processing unstructured data



PRO TIP: Most analytics roles won't require heavy programming unless you work in Data Science or Machine Learning; focus on Excel + SQL first, and build from there

* Python is a general-purpose language (also used for web dev, mobile apps, games, etc.), while R and SAS are geared towards statistical analysis

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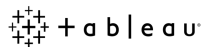
COMMON TOOL STACKS

As you begin to develop your skills, it's important to identify a **collection (or "stack") of tools** which will help you perform the responsibilities of your specific role

BI ANALYST



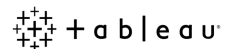
DATA VIZ SPECIALIST



DATA ENGINEER



DATA SCIENTIST



There's no "right" or "wrong" stack, as long as the tools you use allow you to perform your job effectively

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KEY TAKEAWAYS



Think about **skills first, tools second**

- *Don't focus on which tool you want to learn; focus on which SKILL you want to build*



Don't be afraid to **specialize**

- *Aim to go deep rather than broad, and focus on building 1-2 expert-level skills (vs. 10 mediocre ones)*



During the learning process, focus on **one tool at a time**

- *Staying focused helps you learn efficiently, retain your skills, and transfer your knowledge to similar tools*



Work towards building a **“tool stack”** specific to your role

- *For analytics professionals, we recommend Excel, SQL, and a BI platform like Power BI/Tableau*

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TIPS FOR SUCCESS



Obsess over
OUTCOMES



Master **SKILLS**,
not tools



Don't be afraid to
SPECIALIZE



Don't ignore the
SOFT SKILLS



Never stop
LEARNING



Follow your **OWN**
PATH

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